YIXIN WANG 王亿鑫

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Yixin now is a Research Associate at PolyU after receiving his M.S. degree in Management Science and Engineering from SZU. Thanks for reading. 😁😁😁

PUBLICATIONS

JOURNAL PAPERS:

[1] Hong Wang (*Advisor*), *Yixin Wang*, Menglong Liu, et al. "An enhanced bacterial colony optimization with dynamic multi-leader co-evolution for multiobjective optimization problems" ***Expert Systems***, 2023. (**JCR Q2, IF: 3.30**) (Link)(PDF)

[2] Hong Wang (*Advisor*), Yikun Ou, *Yixin Wang*, et al. "Semisupervised bacterial heuristic feature selection algorithm for high-dimensional classification with missing labels" ***International Journal of Intelligent Systems***, 2023. (**JCR Q1, IF: 8.993**) (Link)(PDF)

[3] Shuang Geng, Xiaofu He, *Yixin Wang*, et al. "Multicriteria recommendation based on bacterial foraging optimization." ***International Journal of Intelligent Systems***, 2021. (**JCR Q1, IF: 8.993**) (Link) (PDF)

[4] Hong Wang (*Advisor*), *Yixin Wang*, Sili Wen, et al. "Multiobjective queueing service network optimization for resource allocation in a hierarchical medical system." (*Under review*) (Slides)

[5] Ben Niu (*Advisor*), Bowen Xue, *Yixin Wang**(*Corresponding*). "Spectral analysis and optimization of thermal emitter in thermophotovoltaic technology via NSGA-II." (*Under review*)

CONFERENCE PAPERS:

[1] Hong Wang (*Advisor*), *Yixin Wang*, Mengjie Wan, et al. "Multi-objective bacterial colony optimization based on multi-subsystem for environmental economic dispatching." In ***ICSI*** 2022. **(Best Paper) (EI)** (Link) (PDF)

[2] Hong Wang (*Advisor*), Yikun Ou, and *Yixin Wang**(*Corresponding*). "A multi-objective structure variant bacterial heuristic feature selection method in high-dimensional data classification." In ***DMBD***, 2021. **(EI)** (Link) (PDF)

[3] Hong Wang (*Advisor*), *Yixin Wang*, Yikun Ou*, et al. "Bacterial foraging optimization with leader selection strategy for bi-objective optimization." In ***ICSI***, 2021. **(EI)** (Link) (PDF)

[4] Hong Wang (*Advisor*), Zhuo Zhou, *Yixin Wang*, et al. "Feature selection for image classification based on bacterial colony optimization." In ***ICSI***, 2021. **(EI)** (Link) (PDF)

RESEARCH EXPERIENCES

1. Deep Reinforcement Learning and Operations Management

RESEARCH ASSOCIATE PROJECT IN POLYU (2023)

PROJECT BRIEF:

- Inventory management is a complex sequential decision-making problem with multiple *stochastic natures*, e.g., demand uncertainty, lead time uncertainty, etc., and it is difficult to obtain an analytical solution with exact methods. Leveraging the powerful capabilities of feedback learning in RL and the approximation in DL, we're exploring to train an automated decision-making agent for inventory management, expecting to *bring DL and stochastic optimization together*.

PERSONAL KEY WORKS (ON-GOING):

- Methodologies exploration (PPO, Generative models);

- Coding and experiments.

2. A Fair Simulation Optimisation Framework for Disruption Management in Public Transit

RESEARCH ASSOCIATE PROJECT IN POLYU (2023)

PROJECT BRIEF:

- Rail transport system operation management, e.g., Hong Kong Tramways, is an intractable issue facing various considerations. To enhance the anti-disruption ability and address the fairness issues in such system, we're building a *discrete event simulation framework* and *real time rescheduling optimization models*.

PERSONAL KEY WORKS (ON-GOING):

- Discrete event simulation framework construction (Simpy);
- Real time task generation for motorman via DRL (Monte-Carlo tree searching).

3. Scientific Management of Living Supplies During the COVID-19 Pandemic

PROJECT LEADER AND ORGANIZER (2022)

KEY ACCOMPLISHMENT:

- We solve a *living supplies distribution problem* during the Covid-19 pandemic. The number of multi-echelon living supplies distribution spots and their locations are estimated and determined based on regional geographic and population information of Changchun city. Further, an intelligent agent is trained based on *offline RL* for the number of supplies issued. (Brief Intro)
- **"Optics Valley of China · HUAWEI CUP" Second Prize, Postgraduate group, Rank: 290/4021**.

PERSONAL KEY WORKS:

- Funding application;
- Intelligent decision-making agent construction and training based on BCQ;
- Identification of key issues, Coding, Experiments, Visualization, Report writing.

4. Optimal Design of Thermal Emitter in Thermophotovoltaic Technology

PROJECT LEADER AND ORGANIZER (2021)

KEY ACCOMPLISHMENT:

- This is a *materials combination optimization problem* to satisfy design requirements. The relationship between the emission spectra and material properties of single- and multi- layer are studied respectively. We give the optimal design solution under different emission spectrum requirements via *constructing, solving corresponding optimization models*. (Brief Intro)
- **APMCM Second Prize, Postgraduate group**.

PERSONAL KEY WORKS:

- Implementation and solving for single- and multi- layer structure optimization models;
- Identification of key issues, Coding, Experiments, Visualization, Report writing.

5. Identification and Screening of Anti-Breast Cancer Drug Candidates

PROJECT LEADER AND ORGANIZER (2021)

KEY ACCOMPLISHMENT:

- We provide a stable solution for drugs candidates identification that simultaneously optimizes both bioactivity and ADMET properties. Based on the information provided for ERα antagonists, *quantitative prediction model* for compound bioactivity and *categorical prediction model* for ADMET properties are constructed, respectively. (Brief Intro)
- **"HUAWEI CUP" Third Prize, Postgraduate group.**

PERSONAL KEY WORKS:

- Data preprocessing, Feature fusion predication model construction;
- Identification of key issues, Coding, Experiments, Visualization, Report writing.

Skills

Computer skills:

- **Programming Language**: **Python** Object-Oriented Programming; **MATLAB**;
- **Pytorch** Deep Learning implementation framework and **common DS frameworks**;
- **Git/Github** Project management; **Linux/Shell** OS & Batch script;
- **SPSS** Statistical analysis; **Latex** Writing; …
- **Good at and willing to explore new computer-aided techniques, technology enthusiasts.**

Qualifications & Certificates: Trainee Industrial Engineer, Anonymous Reviewer of ICSI

Selected Honors

- **Excellent Postgraduate Scholarship, Second Prize by SZU (2022)**
- **Excellent Postgraduate Scholarship, Second Prize by SZU (2021)**
- **DGUT Outstanding Graduate (2020)**
- **National Encouragement Scholarship, by Ministry of Education (2017, 2018, 2019)**
- **The KangHua Scholarship, by DGUT (2019)**
- **The GuangDa Scholarship, by DGUT (2018)**
- **Excellent Undergraduate Scholarship, First Prize by DGUT (2017)**
- **Advanced Outstanding Student, by DGUT (2017)**

Miscellaneous

Look forward to communicating with you further~ 😁

Thank you for checking my resume~ 😁